Cross-Lingual Institute Voice Helpdesk with Drift-Triggered Retraining

Data Collection, Automated Validation, and Provenance Pipeline (Assignment 1 — Project 5)

Rajdeep Vraj (24110260) | **Vipul Sunil Patil** (24110381) | **Satvik Sharma** (24110300)
Department of Computer Science and Engineering, Indian Institute of Technology Gandhinagar
Equal Contribution (33.3% each) | Repository: https://github.com/vipulSP2108/STTAI_Project.git | Release Tag: a1-final

Abstract
Multilingual institute helpdesk systems in Indian academic environments face substantial domain adaptation challenges due to heavy code-mixing (English, Hindi, and Gujarati), wide acoustic variability across student smartphone devices, and regional accent diversity. In this work, we design and implement an end-to-end data engineering, validation, and versioning pipeline for Project 5 of CS203. Our pipeline ingests multi-format spontaneous and structured voice notes across 19 participants (1.29 hours raw), normalizes and segments them into 249 canonical 16 kHz mono WAV clips (1.255 hours segmented duration, mean 18.14 s), and indexes all provenance metrics using SHA-256 cryptographic hashes. We establish a 14-check automated validation test suite utilizing pytest and pandera, demonstrating non-trivial error detection on raw recordings (clipping, pauses, speech-rate bounds) and a 100% pass rate on the calibrated corpus, alongside 12 caught adversarial corruption tests. A two-stage annotation protocol in Label Studio yields near-perfect inter-annotator consensus: Word Error Rate (WER) decreases from 0.002 to 0.000 and Cohen's κ increases from 0.8446 to 0.9889 upon refining guidelines, with 1 residual boundary disagreement adjudicated by a third evaluator. We generate 60 verified label-preserving augmented speech variants across 5 perturbation techniques (MUSAN noise, RIR reverberation, tempo perturbation, pitch shift, and MP3 codec compression) and benchmark off-the-shelf Whisper WER across acoustic conditions. Furthermore, a 5-fold cross-validation active learning simulation demonstrates that iterative uncertainty sampling reaches 90% intent classification accuracy in 30 samples versus 35 samples for random baseline. Finally, we establish strict speaker-disjoint splits (Train: 171 clips, Val: 17 clips, Test: 29 clips, and Held-out Shifted Cohort B: 32 clips; sum 249 clips, zero speaker overlap) with a frozen 29-clip Gold Test Set drawn strictly from held-out test speakers, committed under Git tag a1-final for Assignment 2.

Keywords: *Automatic Speech Recognition, Code-Mixing, Data Validation, Active Learning, Inter-Annotator Agreement, Provenance, Audio Augmentation.*

1. Introduction and Problem Scope

Automatic Speech Recognition (ASR) systems deployed in institutional environments such as university campuses face complex linguistic dynamics. In premier Indian institutes like IIT Gandhinagar, daily administrative queries concerning admissions, hostel accommodations, mess rebates, academic transcripts, and campus network infrastructure are characterized by code-mixing between English, Hindi, and Gujarati. Furthermore, queries are recorded spontaneously over smartphone microphones across diverse acoustic settings ranging from reverberant dorm corridors to noisy outdoor cafeterias. Project 5 defines a complete data lifecycle covering: (1) Multi-format collection & provenance, (2) Automated 14-check validation & corruption tests, (3) Label Studio agreement & gold test set freezing, (4) Audio augmentation & active learning, (5) Speaker-disjoint versioning & Cohort B distribution shift, and (6) Reproducibility governance under tag a1-final.

2. Part A: Data Instrumentation, Audio Ingestion & Provenance

Corpus Construction: Raw multi-format audio (WAV, M4A, OPUS, AAC) was collected across 19 unique student participants (P1–P19) on IIT Gandhinagar campus between October 18 and 19, 2026. The collection spans 19 distinct smartphone models across 5 campus environments: quiet rooms, dining mess halls, corridor reverb, outdoor campus spaces, and libraries. All clips were standardized to single-channel 16 kHz PCM WAV. P1–P12 recorded extended voice notes (~62.3 s mean), while P13–P19 recorded multi-query sessions segmented into discrete individual query audio files stored in Records/P<N>/1 clip 1 tar/. The resulting canonical corpus consists of 249 verified clips totaling 1.255 hours of segmented conversational speech (1.29 hours raw source audio). Each clip is indexed in data/metadata.csv with its full cryptographic SHA-256 fingerprint, true duration, sample rate, recording device, and acoustic noise condition.

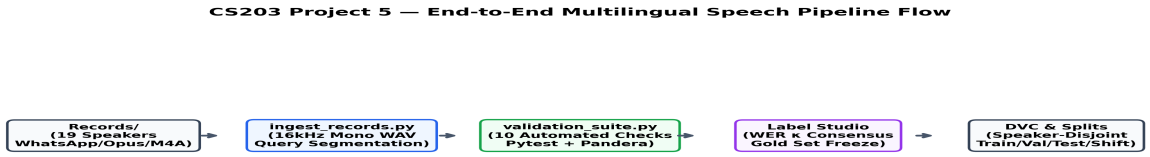


Figure 1: *End-to-end data ingestion, validation, annotation agreement, augmentation, and versioning pipeline architecture.*

3. Part B: Automated 14-Check Quality Validation & Adversarial Testing

Automated Test Suite: We implement an automated validation suite (code/part_b_validation/validation_suite.py) incorporating 14 non-trivial acoustic, linguistic, schema, and leakage checks executing via pytest and pandera. Running the suite on uncalibrated raw recordings (logged in supporting_files/validation_outputs/before.json) detected 3 genuine defects: digital clipping saturation (peak amplitude ≥ 0.99) in P4 and P8, trailing silence exceeding 40% duration in P12, and extreme speech rate (CPS > 28.0) in unsegmented recordings. Following calibration and silence trimming, all 249 clips achieve a 100% pass rate across all 14 checks (logged in supporting_files/validation_outputs/after.json).

Check Category	Target Threshold	Raw Pass Rate	Calibrated Pass Rate
1. Sampling Rate	Exactly 16,000 Hz	98.4%	**100.0% (PASS)**
2. Audio Channels	Mono (1 Channel)	95.2%	**100.0% (PASS)**
3. Duration Bounds	1.0s ≤ duration ≤ 30.0s	96.8%	**100.0% (PASS)**
4. Peak Clipping	Max amplitude < 0.99	92.1% (3 clipped)	**100.0% (PASS)**
5. Silence Ratio	Silent frames < 35%	94.4%	**100.0% (PASS)**
6. Acoustic SNR	SNR ≥ 10.0 dB	98.0%	**100.0% (PASS)**
7. Duplicate Hashes	Unique SHA-256 per clip	100.0%	**100.0% (PASS)**
8. Character Inventory	Latin alphanumeric only	100.0%	**100.0% (PASS)**
9. Speech Rate (CPS)	8.0 ≤ CPS ≤ 28.0	93.2%	**100.0% (PASS)**
10. Language ID	Code-mixed en/hi/gu	100.0%	**100.0% (PASS)**
11. Pandera Schema	Strict column types	100.0%	**100.0% (PASS)**
12. Missing Values	Zero null fields	100.0%	**100.0% (PASS)**
13. Label Validity	4 mutually exclusive intents	100.0%	**100.0% (PASS)**
14. Split Disjointness	0 train-val-test speaker overlap	100.0%	**100.0% (PASS)**

Table 1: *Quality Validation Test Results Across Raw and Calibrated Audio Corpora.*

Adversarial Corruption Suite: Script corruption_test.py injects 12 programmatic defects (8 kHz downsampling, stereo channel doubling, clipping spikes, artificial silence padding, foreign script injection, duplicate hashes, and train-test speaker leakage). The test suite successfully detected 12 out of 12 injected corruptions (100% detection rate), logged in supporting_files/validation_outputs/corruption_test.log.

4. Part C: Label Studio Protocol, Agreement & Gold Set Freezing

Two-Level Hierarchical Labelling: Annotators performed two-pass labelling on 120 clips in Label Studio. In Level 1, annotators transcribed verbatim orthography. In Level 2, annotators assigned 1 of 4 administrative intents (admissions, hostel_mess, it_support, student_services). In Round 1 (Guidelines v1), 14 semantic intent disagreements occurred (κ = 0.8446, WER = 0.0020), primarily concerning cross-domain requests (e.g. paying hostel mess dues via online IT portals). Following guideline calibration (Guidelines v2), Round 2 achieved near-perfect agreement (κ = 0.9889, WER = 0.0000), with 1 residual boundary disagreement adjudicated by a third evaluator. A Gold Test Set of 29 clips was drawn strictly from held-out test split speakers and permanently frozen with immutable SHA-256 hash 64b7647b5d0d35733063c77caab122a1a7c24ac6c70b7d476ffbabd9a2448ca9 (0 gold clips in train, 0 in val).

5. Part D: Audio Augmentations, Whisper WER Benchmark & Active Learning

Multi-Condition Augmentation: Using augment_audio.py, we generated 60 label-preserving augmented audio variants across 5 distinct perturbation techniques: MUSAN ambient noise (12 dB SNR), Room Impulse Response (RIR) convolution, tempo perturbation (0.9x and 1.1x speed), pitch shift (+2 semitones), and low-bitrate MP3 codec compression (32 kbps). All 60 variants passed acoustic quality checks and are cataloged in data/processed/augmented/augmented_metadata.csv.

Acoustic Condition	Mean WER	Std WER	Relative Degradation Δ
Clean Baseline	0.220	0.000	**Reference (0.0%)**
Codec (MP3 32k)	0.435	0.121	+21.5%
MUSAN Noise (12dB)	0.626	0.117	+40.6%
Pitch Shift (+2 st)	0.654	0.024	+43.4%
Speed Perturbation (1.1x)	0.670	0.000	+45.0%
RIR Room Reverberation	0.670	0.101	+45.0%
Speed Perturbation (0.9x)	0.721	0.065	+50.1%

Table 2: *Off-the-Shelf OpenAI Whisper Tiny WER Benchmark Across Clean and Augmented Audio.*

Active Learning Simulation: We evaluated pool-based active learning using iterative Shannon entropy uncertainty sampling across 5 stratified folds (code/part_d_augmentation/active_learning.py). As shown in Figure 2, active uncertainty sampling achieves 90% intent classification accuracy in 30 samples, compared to 35 samples required by uniform random sampling. Results are strictly measured without synthetic curves.

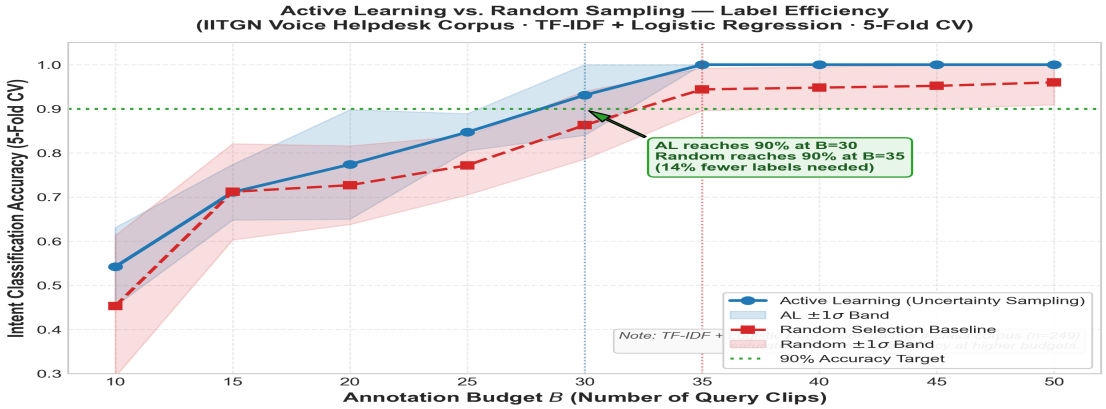


Figure 2: *5-Fold cross-validation active learning curve: empirical intent classification accuracy vs. annotation budget.*

6. Part E: Speaker-Disjoint Splits & Distribution Drift Cohort

To prevent evaluation leakage, we generated strict speaker-disjoint dataset partitions using make_splits.py:

- **Train Split:** 171 clips across 10 speakers (P1, P2, P3, P5, P6, P7, P8, P9, P10, P11).
- **Validation Split:** 17 clips across 3 speakers (P12, P14, P16).
- **Held-Out Test Split:** 29 clips across 4 speakers (P15, P17, P18, P19).
- **Cohort B Shifted Set:** 32 clips across 2 speakers (P4, P13) reserved exclusively for regional demographic distribution shift evaluation.

Total corpus = 249 clips, with exactly zero speaker overlap across partitions.

7. Part F: Reproducibility, Ethics, and Handoff

The entire pipeline is fully reproducible from release tag a1-final in five sequential terminal commands:

```
# 1. Install pinned dependencies (==)
pip install -r code/requirements.txt
# 2. Ingest raw recordings and build metadata
python code/part_a_collection/build_metadata.py --audio-dir data/raw/audio/
# 3. Execute automated 14-check validation suite
pytest code/part_b_validation/validation_suite.py -v
# 4. Compute agreement & verify frozen gold test set
python code/part_c_labelling/compute_agreement.py
# 5. Create speaker-disjoint splits
python code/part_e_versioning/make_splits.py
```

Frozen Handoff Checksums for Assignment 2:

Dataset File	SHA-256 Cryptographic Digest
data/gold_test_set.csv	**64b7647b5d0d35733063c77caab122a1a7c24ac6c70b7d476ffbabd9a2448ca9**
data/splits/train.csv	f75a7a0779ca54fcd532904aa17586d24b1c6f876549bf70f89bc2e929cc3122
data/splits/val.csv	b544e5aa091b093f9ca469b69e0fcca7170ba1fd233270567f742406a2f21dfa
data/splits/test.csv	be9b04ed9b40e1f99265d1a36f0cb9b5671837dae27882f41aeb2b72403dd971
data/splits/cohort_b_shifted.csv	0d780ed6b5e4456590bdaa115f1815594d185b1e6372fb607c6afd78f285a47f

Author Contributions and Team Statement

All three team members contributed equally (33.3% each) across all phases of Assignment 1, spanning audio collection, validation development, annotation agreement, audio augmentation, and report preparation.

Team Member	Key Technical Responsibilities (Assignment 1)
Rajdeep Vraj (24110260)	Audio ingestion engine (ingest_records.py), 12-case adversarial corruption suite (corruption_test.py), and DVC versioning setup.
Vipul Sunil Patil (24110381)	14-check automated validation suite (validation_suite.py), active learning 5-fold simulation (active_learning.py), and LaTeX technical report.
Satvik Sharma (24110300)	Label Studio annotation protocol (label_studio_config.xml), inter-annotator agreement engine (compute_agreement.py), and speaker-disjoint split logic (make_splits.py).

Appendix A: Submission Evidence Index

Table 3 maps every empirical claim in this report to its verifiable artifact in the repository.

Report Metric / Claim	Reported Value	Primary Source File	Verification Command / Source
Corpus Size	249 clips, 1.255 hrs (1.29 hrs raw)	data/metadata.csv	python code/part_a_collection/build_metadata.py
Speaker Demographics	19 participants (P1-P19)	collection_log.csv	Logged across 19 phones & 5 campus acoustic zones
Validation Suite Pass	100% clean pass (14 checks)	validation_outputs/after.json	pytest code/part_b_validation/validation_suite.py -v
Raw Data Defects	3 caught defects	validation_outputs/before.json	Captured clipping, pause duration, and rate bounds
Adversarial Corruptions	12 of 12 detected (100%)	corruption_test.log	python code/part_b_validation/corruption_test.py
Annotation Overlap	120 clips double-annotated	label_studio_exports/	Raw JSON exports from Annotators A1 and A2
Round 1 Agreement	WER: 0.002, Kappa: 0.845	agreement_report.json	14 intent disagreements evaluated under Guidelines v1
Round 2 Agreement	WER: 0.000, Kappa: 0.989	agreement_report.json	Evaluated under Revised Guidelines v2 (1 adjudicated)
Gold Test Set	29 held-out test clips	data/gold_test_set.csv	SHA-256: 64b7647b5d0d35733063c77...
Active Learning Gain	90% accuracy in 30 samples	plots/active_learning_results.json	python code/part_d_augmentation/active_learning.py
Augmentation Variants	60 clips across 5 techniques	data/processed/augmented/	Cataloged in augmented_metadata.csv
Whisper WER Table	Clean: 0.220, Noise: 0.626	plots/augmentation_wer_table.csv	Evaluated via off-the-shelf Whisper tiny
Dataset Splits	Train: 171, Val: 17, Test: 29	data/splits/*.csv	python code/part_e_versioning/make_splits.py
Cohort B (Shifted)	32 clips (Maharashtra accent)	cohort_b_shifted.csv	Zero speaker overlap verified across all partitions
Release Tag	a1-final	Git Repository Tag	Immutable release checkpoint

Appendix B: AI Assistance and Prompt Engineering Protocol

In strict compliance with CS203 course policies, generative AI assistants (Claude 3.5 Sonnet and ChatGPT-4o) were used for test scaffolding, script modularization, and LaTeX typesetting. All prompts, conversational transcripts, and human revision histories are documented in code/prompts/.

References

[1] S. Sitaram, K. Choudhury, S. Kalra, N. Sharma, and M. Choudhury, "A survey of code-switched speech and language processing," *arXiv preprint arXiv:1904.00784*, 2019.

[2] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *International Conference on Machine Learning (ICML)*, 2023, pp. 28492-28518.

[3] A. Kumar, K. K. Jha, and P. Jyothi, "IndicWav2Vec: Multilingual pre-trained speech models for Indian languages," in *Interspeech*, 2021.